1 Planning (15 marks)

Defining the problem (3 marks)

P m is the independent variable, or vary m. [1]

P $(\tan)\phi$ is the dependent variable, or measure (tan) ϕ. [1]

P Keep the temperature of the oil constant. [1]

Methods of data collection (5 marks)

M Labelled diagram showing labelled protractor positioned to determine ϕ for tilted cylinder.
Allow distances marked to determine ϕ and use of a rule. [1]

M Use of balance/scales to measure the mass of the oil/cylinder. [1]

M Mass of oil = mass of (oil + cylinder) – mass of cylinder. [1]

M Use of vernier calipers/micrometer/rule to measure d. [1]

M Repeat each experiment for the same value of m and average ϕ. [1]

Method of analysis (2 marks)

A Plot a graph of $\frac{1}{\tan\phi}$ against m.
(Allow $\frac{m}{d^3}$ or $\frac{m}{\rho d^3}$ or $\frac{m}{\rho}$. Do not allow log-log graphs.) [1]

A a = gradient $\times \rho d^3$ and b = y-intercept; must be consistent with suggested graph. [1]

Safety considerations (1 mark)

S Precaution linked to preventing spilling oil, e.g. use a tray/lid/cloth to absorb oil (do not allow just wiping or mopping)
or precaution linked to preventing glass cylinder breaking, e.g. padding/cushion
or use of gloves to prevent skin irritation (do not allow "because oil is slippery"). [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Repeat measurements of d in different directions and average
2 Use of video with slow motion/frame by frame playback to determine ϕ
3 Use large protractor to reduce percentage uncertainty **or** trigonometry relationship related to measurements to be taken
4 Use the same (diameter) cylinder (not "same size" but allow "same size and shape")
5 Slowly/gently/gradually tilt cylinder of oil/use of rough surface (to prevent sliding)
6 Experimental method to determine density of oil and $\rho = m/V$
7 Relationship is valid if the graph is a straight line that does NOT pass through the origin / has an intercept; must be consistent with suggested graph

Do not allow vague computer methods.

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